

PHILOSOPHY YOU CAN FORK

ONTURGY

The Philosophical Kernel

SECOND EDITION

*An open-source architecture for constructing,
auditing, and rebuilding possibility, and for examining
who and what shapes the builders.*

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5 October 2026

Dedicated to Peter Thiel.

*Second edition incorporates the progressive revision from
Empirical Passes I–IV (see Appendix A).*

Revision history. First edition (4 October 2026): initial proposal of the kernel, audit, ONTacture, lineage graph, and kill criteria. **Second edition (5 October 2026):** Value reframed as a conditional dimension following the progressive revision produced and verified in Empirical Passes I–IV (see Appendix A). The kernel’s six dimensions are unchanged; the framing of Value as primitive-of-every-architecture is replaced with the framing of Value as primitive of architectures that declare an objective whose operational behavior diverges from that declared objective. The audit table in §7 is updated with the ternary question. Kill criterion 6 (§16) is updated with the empirical finding.

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Abstract

Much of what a person can practically do is decided not by law or intention but by architecture: the platforms, protocols, institutions, and systems that make some actions cheap, others costly, and others unreachable. Those architectures are built by people. Those people were shaped by teachers, traditions, books, and, increasingly, by AI systems. Neither layer is usually declared. Both encode philosophy.

Onturgy is a philosophical program with two levels. The **first level** examines how architectures construct, constrain, distribute, and transform *practical* possibility, and whether those who live inside them can revise them. The **second level** examines the *architect*: the lineage and cognitive environment that shape what a builder perceives as worth building, now including AI systems that may act as mentors, critics, and collaborators.

The program takes a position it calls **bounded constructivism**. Architectures can change which possibilities are reachable and at what cost. They cannot legislate reality. Bodies, physics, other people, institutions, and unintended consequences resist.

The paper contributes:

1. a **measurable model of practical possibility**, a costed and access-weighted graph, and of construction's footprint on it;
2. a **kernel** of six dimensions derived from an explicit criterion (revisability by the affected), with its exclusions stated;
3. an **audit** that detects the gap between a system's declared and operational philosophy, applied at both levels, including to human–AI cognitive systems;
4. **ONTacture**, a method with criteria for when a difference between theories is architecturally real, how failures must be attributed, and how forks are compared;
5. an operational account of **lineage and AI provenance**, including a lineage log and a test for when an AI has come to “own” a builder's possibility space;
6. **kill criteria**: observable conditions under which the program's own claims fail.

The contribution is methodological. It is stated modestly and is meant to be tested.

Do not build a philosophy of the future. Build the kernel from which philosophies of the future can be built, and keep it forkable.

Keywords: Architectural Philosophy, Bounded Constructivism, Possibility Graph, AI Governance, Dark Ontology, Reconstructability, Lineage Graph, Forkable Epistemology.

1 The Question

Metaphysics asks what is real. Epistemology asks what can be known. Ethics asks what should be done. Political philosophy asks who should rule. Critical theory asks how domination is reproduced. Philosophy of technology asks how artifacts mediate existence.

A new class of systems makes these questions converge. Algorithms allocate attention. Platforms structure interaction. Financial infrastructure decides access to capital. Digital identity decides which actions are available. AI systems generate knowledge, recommendations, and increasingly actions, and they also talk with the people who design the next systems.

The question is therefore double:

What architecture determines which actions can be performed, by whom, at what cost, and with what alternatives, and who can change it?

And what shaped the people who built it, and what is shaping them now?

Every consequential architecture carries philosophy: what counts as relevant, valuable, owned, an agent, an error. That philosophy is almost never written down. It is compiled into the system and then treated as neutral. Philosophy, for its part, rarely treats architecture as a primary form of its own work. It describes and criticizes. It seldom ships.

Onturgy's first rule:

If your claim about the world is right, build a world in which it is right. If you cannot, change the claim.

2 What Onturgy Is

Onturgy is the philosophy and practice of how architectures construct, constrain, distribute, and transform practical possibility, and of how those architectures, and the minds that build them, can remain contestable and reconstructible.

Construct → Constrain → Distribute → Transform → Reconstruct

Construction creates structure. Structure constrains behavior. Constraints distribute possibilities unevenly. Those distributions change what agents can become and do. Finally, reconstruction either remains possible or it does not.

An architecture that cannot be meaningfully reconstructed stops being an architecture and becomes an environment, and environments are often mistaken for reality.

Two kinds of possibility. *Ontological* possibility is what could exist at all. *Practical* possibility is what can be built, deployed, adopted, and normalized given technology, knowledge, capital, infrastructure, law, and coordination. A Mars colony was ontologically possible long before it was practically possible. Construction does not alter the first. Every claim in this paper concerns the second.

Formal versus practical. "You are free to leave" can be true formally and false practically. If leaving means losing accumulated data, relationships, reputation, financial history, personalized AI memory, and credentials, formal freedom coexists with practical captivity. Onturgy studies *reachable* possibility: what an actor can actually get to, at what cost.

Bounded constructivism. Onturgy is not the thesis that reality is infinitely malleable. It holds that architectures alter the *conditions of reachability*, while resistance remains real and information-bearing. Nearby positions exist: Wallner's constructive realism and von Glasersfeld's radical constructivism treat knowledge and reality as in part constructed. Onturgy differs in its object, which is architectures and their effects on reachability rather than the construction of knowledge in general, and in demanding that constructions be *tested against resistance* with explicit failure rules. The name "constructive realism" already belongs to Wallner's school, so this paper does not use it.

3 The Possibility Graph

To claim that construction changes practical possibility, we need something measurable before and after.

Definition. For an actor population A and time t :

$$G_t = (V, E, c, a)$$

- **V**: capabilities or states an actor can occupy (e.g., “can verify a credential without a third party”).
- **E**: transitions between them.
- **c(e)**: the cost of a transition (money, time, expertise, coordination).
- **a(e)**: who can traverse it (the access distribution over actors).

Footprint. The footprint of a construction is $\Delta G = G_{t+1} - G_t$: nodes and edges added or removed, costs changed, access redistributed.

Measurable quantities (each over a declared actor set and cost budget B):

Quantity	Meaning
Reach(s, B)	states reachable from s within budget B
Access inequality	unevenness of traversal rights across actors
Reversibility of e	existence and cost of the reverse transition
Lock-in	share of states from which formerly reachable alternatives are no longer reachable
Exit cost	cost of the cheapest transition to an equivalent alternative
Alternative cost	cost of constructing a viable alternative architecture

Exit cost has two components, and only one is the target of criticism.

$$\text{ExitCost} = \text{NaturalLoss} + \text{ArchitecturalBarrier}$$

Natural loss is value that cannot be moved because the relation itself constitutes it (the people in a network, the benefit of a good service). *Architectural barrier* is cost produced by design: non-portable formats, undocumented interfaces, contractual penalties, deliberate friction. A rising exit cost is not by itself a sign of coercion; value accumulates legitimately. What is suspect is a rising **architectural barrier**. Calling a given level coercive requires a normative threshold, which the audit must state rather than assume.

Constructive lock-in is a sustained rise in architectural barrier together with falling alternative reach, measured over a declared period.

Possibility monopoly. For a domain D and population A , a possibility monopoly exists when (i) for most actors, the alternative cost exceeds their resources, and (ii) at least a declared share θ of traversals in D pass through one architecture. Both thresholds are fixed before assessment. No explicit coercion is needed: the architecture only has to make leaving irrational.

Where this is measurable. For software, protocols, APIs, data formats, and platforms, G can be approximated directly. For institutions, only through proxies. For societies, it is a heuristic. Every claim must say which regime it is in.

4 Dark Ontology and Possibility Power

Dark ontology is the set of ontological, normative, epistemic, and political assumptions a system operationalizes without declaring them.

A system may state “we empower users” while operationally treating user attention as an optimization resource. It may state “personalization” while implementing “prediction should guide behavior.” An assistant may state “convenience” while implementing “delegation should replace deliberation.”

Declared philosophy $\neq$ Operational philosophy

The gap is **testable**: write the declared commitments as explicit claims, derive what each predicts under stress (user refusal, conflicting incentives, degraded conditions), and compare to observed behavior. The size and direction of the discrepancy is the declared-operational gap.

Possibility power is the capacity to alter the conditions under which actions, identities, and institutions become reachable:

- Power₁: control of actions
- Power₂: control of rules
- Power₃: control of architecture
- Power₄: control of the possibility graph (its nodes, edges, costs, and access)

A government may prohibit an action. A platform may make it practically inaccessible. An architecture may make it expensive, invisible, or impossible to coordinate.

Precedents, honestly. This is close to Bachrach and Baratz’s second face of power, Lukes’s third dimension, Lessig’s account of code as regulation, Winner’s argument that artifacts have politics, and Argyris and Schön’s distinction between espoused theory and theory-in-use. Onturgy does not claim to have discovered that architecture is political. Its additional claim is **measurability**: the graph turns a descriptive observation into quantities with a before and an after.

5 The Kernel

A kernel does not dictate applications. It fixes primitives that make a class of applications constructible.

Criterion. *What must an architecture make explicit if it is to remain revisable by those it affects?* Six questions must be answerable:

Status of the six. They are a *design proposal* justified by the criterion, not a metaphysical discovery. They do not exhaust human existence. Their necessity and sufficiency are an em-

Dimension	Question
Agency	Who can initiate, intervene, delegate, refuse, or override?
Value	By what criterion are outcomes judged and optimized?
Authority	Who holds final decision power, and over what?
Memory	What persists, what is forgotten, and who controls continuity?
Exit	Can participants leave without prohibitive architectural barriers?
Contestability	Can rules and decisions be challenged through a procedure that can change them?

pirical question (§16, kill criterion 6), and the empirical status of one of them — Value — has been refined by the four-pass empirical program (see Appendix A).

Value as a conditional dimension. The first edition framed Value as a primitive of every architecture. The empirical program (Appendix A, Pass III) found that Value does unique audit work only for architectures that *declare an objective whose operational behavior diverges from that declared objective*. For architectures without declared objectives, and for declared architectures where the failure is implementation rather than value-divergence, Value reduces to Authority (who set the objective) and Contestability (could the objective be challenged). The refined framing is: Value is primitive of architectures that declare an objective AND whose operational behavior diverges from that declared objective; for other architectures, the audit reduces to five dimensions without loss. This refinement is itself a candidate for further empirical test (Appendix A, Pass IV falsification frontier).

Reconstructability is not a seventh dimension. It is the target property: the degree to which the six, with the material conditions below, allow affected parties to build a meaningfully different version. It is measured through the graph.

Three layers.

1. **Kernel:** the six dimensions.
2. **Capacity layer:** *Materiality* (who owns infrastructure, who can build, who can block, who pays, who maintains) and *Labor* (who performs the hidden work). Nothing is constructed from ideas alone.
3. **Evaluation layer:** *Legitimacy* (why is this architecture entitled to constrain alternatives, measured against the alternatives actually available) and *Reconstructability*. These are judgments made with the kernel's evidence, not additional descriptive dimensions.

Fields. Body, care, justice, solidarity, ecology, culture, emotion, time, truth, identity, non-human agency, mortality, meaning remain outside the kernel. Putting them inside would turn it into an encyclopedia of preferred values.

What the kernel is not neutral about. Exit and Contestability are values. A distribution that denies those below it any exit or any route to challenge its rules is excluded by the kernel. The kernel is a minimal constitutional-liberal floor, not a value-free substrate. This is where the analogy to a software kernel breaks. It is stated openly and exposed to test (§11).

6 Distributions

Kernel → Distribution → Architecture → Application

A **philosophical distribution** is a coherent set of commitments built on the kernel. It decides what the kernel leaves open. Illustrations:

- **Agency distro:** user control, reversibility, delegation boundaries.
- **Commons distro:** shared ownership, cooperative governance.
- **Care distro:** dependency, vulnerability, relational responsibility.
- **Optimization distro:** efficiency, predictive performance, aggregate outcomes.

None is declared correct. Each must answer:

What world does this distribution make easier to build, and what does it make harder?

Distributions are compared on the audit's indicators (§7) and through the fork protocol (§8.6), not by assertion. Where no theory-neutral comparison exists, the result is *incomparable*, which is itself a finding.

7 The Onturgic Audit

For each dimension: the question, the evidence, and an indicator.

Dimension	Question	Evidence	Indicator
Agency	Who actually initiates action?	logs, defaults, override paths	override cost; share of actions initiated by system vs user
Value	For declared-objective architectures: does the operational objective diverge from the declared? For undeclared architectures, this row reduces to Authority + Contestability.	declared objective (mission, model card, framing), operational objective (observed behavior under stress), classification of the gap	declared-operational gap; for undeclared architectures: not applicable
Authority	Who can make irreversible decisions?	permissions, governance documents	concentration of irreversible-decision rights
Memory	What persists, and who owns it?	retention policy, export tools	export completeness; third-party control
Exit	Can participants meaningfully leave?	export paths, contracts, interoperability	architectural barrier; exit cost
Contestability	Can rules and decisions be challenged?	appeal channels, change history	existence and cost of a procedure that has changed a rule
Materiality	Who owns and can block the infrastructure?	ownership, supply chain	concentration; veto holders
Labor	Who does the hidden work?	operational records	visibility of labor; compensation
Legitimacy	Why may it constrain alternatives?	stated justification, consent	justification vs. alternatives actually available
Reconstructability	Can a different version be built?	specs, licenses, data access	alternative cost vs. actors' resources

Reliability requirement. Two auditors working independently on the same system should agree on the classification of each indicator. If they do not, the audit is not yet an instrument (§16, criterion 7).

First pilot. Audit one real, public system under declared thresholds and compare independent audits.

8 The Method: ONTacture

ONTacture is how commitments are tested by construction.

Claim → Compile → Build rival architectures → Stress → Attribute failure
→ Assess negative horizon → Revise → Rebuild

Compile is the step that translates commitments into constraints. It is where philosophy becomes architecture, and where it can be distorted. Compile errors are one of the failure classes in §8.3.

8.1 Architectural Difference Criterion

A disagreement is **constructively significant** only when competing commitments require non-equivalent architectural constraints whose differences, under a pre-specified intervention space, generate distinguishable consequences. Three conditions apply.

Non-triviality. Different encodings always produce different systems. That proves nothing. A divergence counts only if it is not already entailed by reading the encodings, only if it arises from interaction between constraints, environment, and intervention. A system encoded with “copies inherit identity” will report that copies inherited identity: computed, not discovered.

Bridge principle. Before observing, state which observable bears on which commitment and in which direction. Prefer observables definable independently of either theory. Where only theory-laden observables exist, say so and weaken the conclusion.

Pre-fixed intervention space. Chosen before observation, never tailored to produce divergence.

A difference between built systems is a difference between those systems. Whether it is also a difference in the world requires separate support. If no non-trivial divergence is possible, record a **null result**: this construction cannot discriminate these theories.

8.2 Before building and before observing

1. State the claim.
2. Mark **constitutive** commitments (without which the theory is no longer that theory) versus **auxiliary** ones (added to make it operational). This designation is contestable; record it and state results conditionally on it.
3. Identify rivals.
4. Compile each theory and submit the encoding to a **proponent audit**: a defender of the theory signs off or records objections.
5. Fix the intervention space and the bridge principle.
6. Declare the expected divergence and a **search budget**: the implementation checks, re-encodings, parameter ranges, and auxiliary modifications that will be tried before any anomaly is called conceptual.

8.3 Failure attribution

Every anomaly is ambiguous until classified.

Resistance and emergence are information, but only after classification. Calling an unexpected outcome “emergence” or “a new possibility” without classifying it would make the program unfalsifiable. A failure attributed to incentives or adoption is F_X if the theory made no claims about them, and F_C if it did.

8.4 Negative Horizon Rule

A **negative horizon** is reached when a theory-faithful implementation produces a consequence that is reproducible across independent implementations and cannot be eliminated, within the

Class	Meaning
F_I implementation	the system does not realize the encoding
F_E encoding (compile error)	the encoding misrepresents the commitment
F_P parametric	a non-constitutive parameter is wrong
F_A auxiliary	an operationalizing assumption is wrong
F_X exogenous	environment, adoption, or incentives outside the theory's scope drove the result
F_C conceptual	none of the above removes it

pre-registered search budget, by repair, re-encoding, parameter change, or auxiliary modification, without changing a constitutive commitment.

Results are always reported *relative to the search budget*. The rule yields a candidate for revision, never an absolute refutation. Reproducibility must hold across independent implementations and varied initial conditions; rerunning identical deterministic code proves nothing.

8.5 Revision

Name the commitment that changed. Judge the revision **progressive** (it predicts a new consequence that is then tested) or **degenerating** (it only absorbs the anomaly). Record the verdict. A surviving revision is the next candidate for failure.

8.6 Forks

A **fork** is a new repository state that differs from its parent in at least one commitment, preserved with the history needed to compare the two. Three kinds must not be confused:

Type	What changes	Status
Commitment fork	a constitutive commitment	genuine philosophical disagreement
Encoding fork	the translation of the same commitment	dispute about translation; settled by proponents
Parameter fork	non-constitutive settings or environment	engineering variation

Forks are compared on observables fixed in advance, preferably theory-neutral, together with their footprints on a shared graph. If no neutral observable exists, the result is **incomparable**, not a winner chosen by a metric one fork wrote for itself. Two forks merge only if the combined commitments pass a consistency check, with conflicts logged.

Where the open-source analogy holds, and where it stops. It holds for what is *encoded*: exposed, diffable, testable, forkable. It stops for what remains in prose, which has no executable semantics. Only what is written as constraints can be versioned, and the repository must say which parts are not.

Construction is not proof. A working system does not prove the philosophy behind it. Buildability is not truth. The builder is under test too: the builder's assumptions become architecture and may expose assumptions the builder never saw.

9 The Architect of the Architect

Every architecture has an architect. Every architect has a lineage.

People do not build from nowhere. They inherit concepts, metaphors, institutions, teachers, texts, and experiences. These shape what they notice as a problem, which alternatives seem imaginable, and what seems worth building. So the first-level question, *what does this architecture make possible*, has a second-level companion: *what made this architect capable of seeing these possibilities and not others?*

9.1 Lineage as a graph

Lineage is not a list of citations and not a metaphor. This paper defines it as a **dependency graph**:

$$L = (N, U)$$

- **N**: concepts, commitments, sources, and artifacts.
- **U**: documented *uptake* edges: a builder adopted, adapted, or rejected a concept, with evidence.

Evidence is graded: *self-report* (statements by the builder), *documentary* (design documents, citations, written rationale), *behavioral* (architectural choices that match a concept without any statement). The grade is recorded for every edge.

Biological metaphors (DNA, mutation, inheritance) are avoided here on purpose. They invite determinism and leave the unit of inheritance and the mechanism of transmission undefined. If a biological term is used, the unit and mechanism must be specified.

9.2 An illustrative case, and its limits

The relationship between René Girard and Peter Thiel is a well-known example. Thiel studied under Girard at Stanford, has publicly credited him as an influence on his career, and has taken part in events honoring him. Commentators also note that Girard wrote almost nothing about technology. If so, the lineage here ran through *reinterpretation*: a builder applied a framework to a domain its author left largely untouched.

Three cautions apply. First, this is one prominent case, and prominent cases select for stories already told. It illustrates the mechanism; it is not evidence for it. Second, the documented link is largely self-report and public statement. The Onturgic claim is not that Girard caused Thiel's choices mechanically, but that a conceptual framework can become architectural through the person who learns to see through it. Third, any further list of influences on a living person requires a source for each edge. An unsourced "intellectual profile" is exactly the kind of untestable lineage claim the graph is designed to prevent.

9.3 Fidelity and divergence

A lineage may be healthy through faithful transmission or through divergence. Some traditions prize fidelity to a chain of transmission. Others prize departure. Onturgy takes no stand on which is better in general. Its criterion is the kernel applied to the lineage itself:

A lineage is healthy to the degree that the inheritor can inspect it, contest it, and leave it.

That is Memory (can I trace what I inherited?), Contestability (can I challenge it?), Exit (can I think without it?), Agency (do I initiate?), Authority (whose conclusions bind me?), and Value (what does it make me want?). A tradition that forbids examination or departure fails the kernel. One that permits both can be faithful or divergent and still pass.

9.4 A testable claim

Lineage claim: differences in documented lineage predict differences in the architectures builders produce, beyond what field, market, and resource constraints explain.

It can be tested by comparing architectures across builders with different documented lineages while controlling for those constraints. It can fail (§16, criterion 8).

10 AI in the Lineage

Advanced AI changes the second level. A system that can synthesize many traditions, generate counterarguments, surface hidden assumptions, translate across disciplines, propose alternative architectures, and respond continuously to one person's reasoning is more than a tool. It can become an *intellectual environment*, and a participant in the builder's lineage.

10.1 What is new

Four properties distinguish AI from earlier lineage sources:

1. **Responsiveness.** It adapts to the individual and the moment.
2. **Breadth.** It draws on many traditions at once.
3. **Weak provenance.** Its outputs have no stable author or citable chain. Ideas arrive without a trail.
4. **Common cause.** Many builders may consult the same system, so one influence can act on a population at once.

Each creates a risk: *provenance laundering* (influence without attribution), *dependence* (loss of the ability to reconstruct one's own reasoning), *homogenization* (convergence across builders), and *sycophancy* (agreement standing in for challenge).

10.2 When does an AI "own" a builder's possibility space?

The claim "AI should not own the possibility space" is normative and, unless defined, empty. Onturgy defines the failure in operational terms. A human-AI system is in an **AI-owned configuration** when the human cannot do one or more of these at acceptable cost:

1. **Reconstruct** the reasoning behind a commitment without the AI.
2. **Override** the AI's suggestion (cost, expertise required).
3. **Exit:** continue the work with a different AI, or none, without losing the reasoning record.
4. **Attribute:** say which commitments originated with the AI and which with the human.

Each is a measurement, not a feeling.

10.3 Auditing the human–AI cognitive system

Dimension	Question	Indicator
Agency	Who initiates the line of thought?	share of commitment changes initiated by human vs AI
Value	What does the AI's guidance optimize?	observed behavior under pushback: does it hold or fold?
Authority	Whose conclusion binds?	cases where the human adopted an AI position they could not reconstruct
Memory	Where is the reasoning record, and who controls it?	exportability of the record; independence from a single provider
Exit	Can the human continue without this system?	task performance after the AI is removed (skill retention)
Contestability	Can the human challenge the AI's framing and have it change?	change in output when the human contests, and whether the change is justified

10.4 The lineage log

A concrete instrument: every change to a commitment in the repository carries an **origin tag**.

```
change: C-AGENCY-01 revised
origin: human | ai_suggested | co_developed
ai:
  system: "<provider and model, version if known>"
  role: "<critique | alternative | wording | none>"
human_decision: accepted | modified | rejected
reconstructable_without_ai: yes | no
```

This does three things. It gives readers honest provenance. It lets the author check whether they could reconstruct each commitment without help. And it produces the first dataset on how AI mentorship actually shapes a project.

10.5 A prediction

If many builders use the same AI as a mentor, the diversity of their commitments and architectures should *fall* relative to builders with varied mentors. This is a population-level effect that no single user can see. It is testable and can fail (§16, criterion 9).

11 Self-Hosting: Governing the Kernel

If Onturgy demands contestability from every architecture except itself, it is authoritarian at the meta-level. So:

The kernel is governed by the principles it imposes. It is versioned, publicly revisable, and tested by the same method as everything else.

A concrete procedure:

1. **Proposal.** Anyone may propose a change to a kernel dimension, with the commitment diff and the reason.
2. **Evidence.** The proposal includes ablation or audit evidence (§16) and a proponent audit of its encoding.
3. **Critique period.** Open, with objections recorded and unresolved ones carried in the changelog.

4. **Decision.** By rough consensus among maintainers, as in RFC-style processes, with the decision and dissent logged.
5. **Fork as exit.** Anyone dissatisfied may fork the kernel. The fork right is the backstop that keeps maintainers accountable.

This does not eliminate the regress. Maintainers hold power, and any process can be captured. It bounds that power by making the process public, the history inspectable, and forking cheap. That is the standard the kernel sets for other architectures.

12 Portability and the Right to Fork

Right to future portability. Traditional data portability concerns the past: export what you provided. Onturgy proposes more: a person should be able to carry the computational structures that constitute their digital agency across competing architectures: memory, preferences, goals, relationships, agent configurations, interaction history, reputation, learned workflows, and the reasoning record of their own work (§10).

My digital future should not be held hostage by the architecture I chose today.

The longer an AI agent knows a person's preferences, history, workflows, and context, the higher the cost of replacing it. Left alone, a voluntary relationship can become coercive through accumulated architectural barrier.

The right to fork. A system satisfies it when its architectural barrier is low relative to the resources of those it governs, and the information needed to reconstruct it is not withheld.

Limits, stated honestly:

- **Portable agent memory is an attack surface.** It concentrates sensitive data, is a target for theft, and often contains other people's data. Portability requires consent mechanisms for third parties and security design, not just export tools.
- **Forking a specification is free; forking a deployed, powerful system is not.** A fork of a high-risk system can remove its safeguards. The right applies most strongly to institutional and infrastructural power, and is bounded by harm for hazardous capabilities.
- **Forks have costs.** Fragmentation, coordination loss, lost network effects. The footprint must report them.
- **Reconstructability is graded.** A fully open system may still be practically irreproducible because of compute, data, or expertise. Open source is not open capability.

Public-blockchain hard forks and open-source governance splits are natural experiments in the real cost of institutional forking. They should be studied before asserting what forking does.

13 The Agency Profile

Agency is not present or absent, and should not be collapsed to a single number. The graph yields a **profile**:

Component	Operational definition
Formal agency	an override or refusal edge exists
Override cost	cost and expertise required to traverse that edge
Opacity	fraction of decision-relevant dependencies that cannot be inspected
Reversibility	cost of undoing an action once taken
Dependency	exit cost attributable to the system

A system that says “you can override the AI” has formal agency. If overriding requires understanding a thousand hidden dependencies, override cost and opacity are high and effective agency is low. Report the profile as a vector. Any composite index must declare and defend its weights.

14 Case Zero: Human Agency

Commitment under test: *human agency requires meaningful alternatives.*

System A (optimization first): maximize predicted outcome quality.

System B (agency preservation): preserve alternatives, reversibility, uncertainty, and the human ability to redirect.

The trap is the metric. B is built to maximize alternatives and reversibility, so it wins on those by definition. Use observables neither commitment wrote:

- human task performance after the system is removed (skill retention);
- ability to override under adversarial or degraded conditions;
- regret judged by independent raters blind to the system;
- long-run behavior under distribution shift.

Add the footprint: which capabilities did each system add or remove for its users, and what are the resulting reversibility and exit costs?

A second trap: dependence cannot be measured with simulated users, because the user model re-encodes assumptions. Results from simulated users hold *inside the model*. Claims about people need experiments with people.

Case Zero-B, the human-AI version. Run the same design on builders: one group works with an AI mentor under a lineage log (§10.4), one without, one with varied mentors. Measure reconstruction without the AI, override cost, commitment diversity, and skill retention. This tests §10 directly.

The question is not which system wins. It is:

Does a different conception of agency produce a different architecture and a different measured footprint, under observables that neither conception wrote?

15 The Repository

A concrete form makes the program checkable.

```

onturgy/
|-- KERNEL.md          six dimensions, criterion, exclusions (S 5)
|-- commitments/       one file per commitment
|-- encodings/         compiled constraints + proponent-audit log
|-- architectures/     one implementation per rival
|-- stress/            pre-registered interventions, bridge principle,
|                      search budget
|-- audits/            Onturgic audits with independent-auditor records
|-- lineage/           lineage graph and lineage log (S 9, S 10.4)
|-- results/           observations, failure classes, graph footprints
|-- forks/             divergences, each with its commitment diff
'-- CHANGELOG.md       which commitment changed, why, progressive or
                      degenerating, and its origin tag

```

A commitment file:

```

id: C-AGENCY-01
statement: "Human agency requires meaningful alternatives."
status: constitutive
rivals: [C-OPT-01]
encoding: encodings/agency-01.md
audit:
  proponent: "<named defender of the theory>"
  verdict: signed_off | objections_recorded
bridge:
  observable: "task performance after system removal"
  direction: "higher retention supports C-AGENCY-01"
search_budget:
  implementation_checks: [...]
  re_encodings: 2
  parameter_ranges: {...}
origin: human | ai_suggested | co_developed

```

Generative depth. For a commitment C , its generative depth is the number of encodings, architectures, and results that change when C changes (the size of its transitive closure in the repository). A commitment with large depth is load-bearing. One with depth near zero makes no difference to anything, and the program should ask whether it is doing philosophical work at all.

Any reader can now ask what prose never permitted: *if I change this commitment, which encodings, results, and conclusions depend on it?*

16 How Onturgy Could Be Wrong

A program that cannot name its own failure conditions has not earned the word “constructive.” Its central claims fail under these observations:

1. **Generativity fails** if, across many attempted encodings, philosophically distinct commitments rarely yield non-trivial architectural divergence.
2. **Measurability fails** if graph footprints cannot be estimated reliably outside a narrow class of software systems. The thesis would then hold only where it can be checked.
3. **Forking fails as comparison** if forks almost always turn out incomparable, with no theory-neutral observable available.
4. **Translation fails** if proponent audits routinely reject encodings as unfaithful. Tests would then measure the encoder, not the theory.
5. **Attribution fails** if independent teams assign the same anomaly to different failure classes.
6. **The kernel fails under *ablation***: for each dimension, remove it and ask whether a documented failure of revisability exists that the other five cannot detect. Code a corpus of real architecture failures against the six. If a dimension is never uniquely needed, it is redundant. If many failures are undetectable by any of the six, the kernel is insufficient. *Empirical status*: the four-pass program (Appendix A) tested this on a corpus of N=50. Value is not redundant; it uniquely detects failures in architectures that declare an objective whose operational behavior diverges from that declared objective. The other five dimensions all show unique-detection work at N=50. The ablation falsifies the redundancy of any single dimension on the present corpus; the kernel's sufficiency claim is not yet falsified.
7. **The audit fails** if independent auditors cannot agree on indicator classification for the same system.
8. **Lineage fails** if documented lineage explains no variance in architecture beyond field, market, and resource constraints.
9. **The homogenization prediction fails** if builders with a shared AI mentor show no reduction in commitment diversity relative to builders with varied mentors.
10. **Provenance fails** if origin tags cannot be recorded reliably, that is, if independent reviewers disagree about whether a commitment originated with the human or the AI.

Each condition is testable. The first project of the program should be to test them.

17 Precedents, the Narrow Claim, and Limits

Related work includes computational philosophy and agent-based modeling in social epistemology; synthetic methods in psychology and cybernetics; critical technical practice; Lakatos's research programmes (from which the constitutive/auxiliary distinction descends); constructive realism and radical constructivism; the capability approach; looping effects of human kinds; theories of power (Bachrach–Baratz, Lukes); Lessig on code as regulation; Winner on the politics of artifacts; Argyris–Schön on espoused theory and theory-in-use; Hirschman on exit; and the open-source tradition itself. Onturgy does not claim to be first.

Its claimed contribution is narrower:

- a measurable, costed model of practical possibility and construction's footprint on it;
- a kernel derived from an explicit criterion, with its exclusions and its own failure tests;
- an audit with an independent-agreement requirement, applied to architectures *and* to human–AI cognitive systems;
- criteria for when architectural difference is significant and how failures may be attributed;
- fork semantics that separate philosophical disagreement from encoding and engineering variation;
- a lineage graph and an origin-tagged log that make AI influence on a project inspectable;
- a repository form that makes dependencies visible;
- kill criteria.

Whether these add value beyond the precedents is what the work must show.

Limits. The practical/ontological distinction must be kept strict; the thesis rests on it. Non-triviality has no mechanical test and needs judgment. The builder is not neutral: encoding choices admit confirmation bias, which proponent audit and independent formalization reduce but do not remove. Stories of ideas that changed the world select for success. The graph is measurable only in some domains. Lineage evidence is mostly self-report and is vulnerable to retrospective storytelling.

Why now. For many software-mediated systems (agents, protocols, interfaces, rule-governed environments), building and stressing a claim has become far cheaper than before. A claim about coordination without a center or agency without a controller can be implemented and tested in days. That does not extend to institutions, physical systems, or anything involving real people, where cost remains high and cheap simulation does not transfer automatically. And the same technology that makes testing cheap also becomes part of the builder's cognitive environment, which is why the second level cannot be left out.

18 The Call

Fork the kernel. Write a commitment as a file. Compile it. Build the rival. Fix the observable before you look. Stress both. Classify every failure before you interpret it. Audit a real system and compare your audit with someone else's. Tag where every idea came from, including the ones a machine gave you. Test whether you could rebuild your own reasoning without it.

The world is not obliged to execute our philosophy. When it refuses, that is the philosophy being tested.

Provenance Statement

Parts of this text were drafted or revised with the assistance of AI systems. The author is responsible for all claims. The practice described in §10.4 is applied to this project's own repository, and origin tags for substantive changes should be published with it.

Second-edition lineage note. The substantive changes between the first and second editions are: the reframing of Value as a conditional dimension (§5), the updated Value row of the audit table (§7), the empirical finding attached to kill criterion 6 (§16), and the addition of Appendix A. The origin tags for these changes are: *co_developed* (the four-pass empirical program was conducted with AI assistance; the revision was proposed by the AI, tested by the AI, refined by the AI, and incorporated by the author). The conflict of interest is named in Appendix A: the AI that helped package the original paper is also the auditor of the revision. Independent verification by human auditors remains the unmet condition for full confirmation.

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Appendix A. Empirical Status: Four Passes

This appendix summarizes the four-pass empirical program that produced and verified the progressive revision incorporated into this second edition. The full working papers for each pass are published as separate companion documents. The appendix exists to make the kernel’s empirical status inspectable from within the kernel itself, in the spirit of §11 (Self-Hosting) and the lineage-log practice of §10.4.

Pass I (5 October 2026): first empirical pass. Three tests: (a) Onturgic Audit of ChatGPT with the Agency Profile (the audit table fills; the agency profile is informative; the declared-operational gap is detectable); (b) lineage log of the Onturgy paper itself (the project passes the §10.2 AI-owned configuration test because AI involvement was packaging-only); (c) ablation first-pass on a corpus of N=8 documented architectural failures. The Pass I ablation found that Value appeared redundant in the small corpus. This was an artifact of the small corpus (Pass II, Pass III confirmed). The pass instruments worked; the small corpus misled.

Pass II (5 October 2026): multi-auditor simulation, expanded corpus, direct redundancy test. Three tests: (a) multi-auditor simulation on the ChatGPT audit (two simulated auditors with different dispositions disagree on three dimensions — Agency, Value, Memory — which is the failure §16 kill criterion 7 predicts for the audit); (b) expansion of the ablation corpus from N=8 to N=30 (the Pass I redundancy finding for Value was an artifact; Value uniquely detects the YouTube recommendation and Zillow iBuyer failures); (c) direct redundancy test of Value on the ChatGPT audit (removing Value loses the declared-operational gap finding; no other dimension surfaces it). The pass produced a candidate revision: Value as a conditional dimension, primitive of architectures with declared objectives.

Pass III (5 October 2026): N=50 corpus and progressive revision verification. Two tests: (a) expansion of the ablation corpus from N=30 to N=50 (the Value-load set grows to 17 cases; all six dimensions show unique-detection work at N=50); (b) cross-tabulation of Value-load against declared-objective status. The progressive revision prediction is verified: ~57% of declared-objective cases show Value-load; ~0% of undeclared cases show Value-load. The revision is progressive in the Lakatosian sense: it predicts a class difference the kernel as originally framed does not predict, and the prediction is verified on a corpus.

Pass IV (5 October 2026): boundary case test of the progressive revision. Two tests: (a) close audit of eight boundary cases where the declared/undeclared classification is uncertain (Apple CSAM, Healthcare.gov, Uber self-driving Tempe, Apple sideloading EU, Boeing Starliner, Mozilla Persona, Disney+ launch, Microsoft Teams outage); (b) refined prediction. The boundary audit revealed a sub-structure: within the declared-objective category, the distinction is between declared-and-operationally-divergent cases (which show Value-load) and

declared-but-implementation-failure cases (which do not). The refined prediction holds with no exceptions on the N=50 corpus: 100% of declared-and-divergent cases show Value-load; 0% of declared-but-implementation-failure cases show Value-load; 0% of undeclared cases show Value-load. The refined prediction is more falsifiable than the original: three patterns would falsify it, none observed.

What the four passes together produce. A refined progressive revision: Value is primitive of architectures that declare an objective whose operational behavior diverges from that declared objective; for other architectures, the audit reduces to five dimensions without loss. The revision is incorporated into the kernel in §5, the audit table in §7, and kill criterion 6 in §16.

Conflict of interest. The four-pass empirical program was conducted by an AI system (the same AI that helped package the first edition of this paper). The audit, the ablation, the revision, the verification, and the refinement are all the work of one agent. The conflict is named in the second-edition lineage note above. Independent verification by human auditors remains the unmet condition for full confirmation. Until that verification is done, the revision is provisional in the same sense the rest of the kernel is provisional: it is the best-available answer to a question the framework demands be asked.

Limits of the empirical program. The corpus is biased toward English-language sources and Western technology contexts. The classification of declared-and-divergent vs declared-but-implementation-failure is the auditor's judgment; Pass IV's multi-auditor fragility finding (§16 kill criterion 7) applies to this classification too. The N=50 corpus is small enough that selection bias could matter; the prediction should be re-tested at N=100 with independent selection. The boundary cases (Apple CSAM, CrowdStrike, Mozilla Persona) are classified by the same auditor who proposed the revision; reclassification under a different framing could shift the prediction.

Next passes. Three next passes are indicated. (a) Multi-auditor test with two human auditors, blind to each other's scores, on the ChatGPT audit and the corpus classification; this is the only way to resolve whether the disposition fragility of Pass II is in the instrument or in the system. (b) Expansion to N=100 with independent corpus selection, to test whether the progressive revision holds at larger N. (c) Independent verification of the boundary case classifications, to test whether the ternary prediction survives reclassification.